MIL-B-52293B(ME)

22 April 1985

SUPERSEDING
MIL-B-52293A(ME)
16 August 1973

MILITARY SPECIFICATION

BRACKET, RAFT: FOR BRIDGE,

FLOATING, PNEUMATIC FLOAT, CLASS 60

This specification is approved for use by the Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of bracket for military rafting operations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5420

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SPECIFICATIONS

FEDERAL

QQ-S-781	- Strapping, Steel and Seals.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-889	- Dissimilar Metals.

2.1.2 Government drawings. The following Government drawings form a part of this specification to the extent specified herein.

DRAWING

ME

C7970-1	- Bracket, Raft: for Bridge, Floating, Pneumatic Float, Class 60.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

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AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI/ASME	- Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.
ANSI/AWS A5.1	- Covered Carbon Steel Arc Welding Electrodes.
ANSI/AWS A5.18	- Carbon Steel Filler Metals for Gas Shielded Arc Welding.
ANSI/AWS D1.1	- Structural Welding Code.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY, INC. (AWS)

AWS B3.0 - Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 550 Northwest Le Jeune Road, Miami, FL 33126.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The raft bracket shall be as shown on Drawing C7970-1 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings in insure correct fit, assembly, and operation of the raft bracket. Any data (e.g.

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shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or the designated representative.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and as shown on the drawing. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The raft bracket shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any glavanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the raft bracket may be newly fabricated from recovered materials to the maximum extent practicable, provided the raft bracket produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the raft bracket.

3.4 Welding electrodes. Welding electrodes shall conform to ANSI/AWS A5.1 or A5.18, classification and size as required.

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3.5 Treatment and painting. The raft bracket shall be treated and painted in accordance with MIL-T-704, type B.

3.6 Identification marking. The raft bracket shall be metal-die stamped as shown on the drawing. Die stamping shall be legible after painting.

3.7 Stenciling. The nomenclature, National stock number, weight, and cubage shall be stenciled on each bracket. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede the cubic displacement. Markings shall be gothic-type capitals and arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Steel fabrication. Steel used in the fabrication of the raft bracket shall be free from laminations, kinks, and sharp bends. The straightening of material shall be done by methods that will not cause injury to the metal. Shearing and chipping shall be done neatly and accurately. Corners shall be square and true. Flame cutting with a tip suitable for the thickness of metal may be employed instead of shearing or sawing. Burned surfaces of flame-cut material shall be ground or machined sufficiently to remove ash and cooling checks. All bends shall be made with metal dies or fixtures to insure uniformity of size and shape. Precautions shall be taken to avoid overheating, and heated metal shall be allowed to cool slowly. All burrs and rough edges shall be ground smooth.

3.8.2 Welders and welding.

3.8.2.1 Welders. Before assigning any welder or automatic welder equipment operator to welding work covered by this specification, the contractor shall obtain certification that the welder or operator has passed qualification tests as prescribed by either of the following listed codes for the materials joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

ANSI/ASME, boiler and pressure Vessel Code, Section IX, Welding Qualification

AWS B3.0, Welding Procedure and Performance Qualification.

Certification shall be made available for review by the contracting officer or designated representitive.

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3.8.2.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc or gas metal-arc (MIG) process and shall conform to the provisions of ANSI/AWS D1.1, Structural Welding Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed, and the deposit and adjoining base metal shall be wire brushed clean.

3.8.2.3 Welding jigs. Components shall be assembled in steel jigs or frames and welded while held in position. The jigs or frames shall be of a design that will limit the distortion to an amount that can be removed without changing the physical characteristics of the metal of the parts being welded.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and drawings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The first article raft bracket shall be examined for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The raft brackets shall be examined as specified herein, for the following defects:

Major

101. Materials not as specified.
102. Materials are not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatment.
105. Used, rebuilt or remanufactured components, pieces, or parts incorporated in the raft bracket.
106. Nonconformance to the dimensions shown on the drawings.
107. Components missing or not as specified.
108. Steel fabrication not as specified.
109. Welder certification not available for review.
110. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metals.
111. Welding jigs not used or not as specified.

Minor

201. Identification marking incorrect, missing, or illegible.
202. Treatment and painting not as specified.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

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4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

112.	Brackets not bundled as specified for level A.
113.	Bundles not secured as specified for level A.
114.	Strapping not as specified for level A.
115.	Wood filler strips not provided as specified for level A.
116.	Marking missing, illegible, incorrect or incomplete for level A or commercial.

5. PACKAGING

5.1 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. The brackets shall be bundled together in quantities of four. The bundles shall be secured with not less than three flat steel straps conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable. Wood filler strips shall be provided as necessary to obtain uniform and compact bundles.

5.1.2 Commercial. Brackets shall be packed in accordance with ASTM D 3951.

5.2 Marking.

5.2.1 Military. Marking for military level of protection (level A) shall be in accordance with MIL-STD-129.

5.2.2 Commercial. Commercial marking shall be in accordance with ASTM D 3951. Bundles shall be marked with the weight and cube data.

6. NOTES

6.1 Intended use. The raft bracket is intended for use in raising and lowering the ramp approaches of rafts to accommodate varying shore conditions.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Degree of packing required (see 5.1).

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6.3 First article. When a first article inspection is required, the item will be examined and should be a first produced raft bracket. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations and approval of the first article.

Custodian:
Army - ME

Preparing activity:
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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-B-52293B(ME)	Bracket. Raft: For Bridge, Floating, Pneumatic Float,Class 60

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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